information

Article

From Retrieval to Faithful Memory Use: Context-Grounded Evidence Chains for Music Education Agents

Jing Mao [1], Yuwen Chen [1,*], Wei Wan [1], Kun Liu [1] and Rui-Feng Wang [2,*]

[1] School of Humanities and Arts, Hunan Institute of Traffic Engineering, Hengyang 421009, China; maojing2025@hnjt.edu.cn (J.M.); wanwei2026@hnjt.edu.cn (W.W.); liukun@hnjt.edu.cn (K.L.)
[2] Department of Crop and Soil Sciences, College of Agriculture and Environmental Sciences, University of Georgia, Tifton, GA 31793, USA
* Correspondence: chenyuwen2025@hnjt.edu.cn (Y.C.); sweefong.wong@uga.edu (R.-F.W.)

Abstract

Artificial intelligence systems are increasingly used for music education and music score understanding, yet most existing systems answer each task in isolation. This single-instance paradigm makes it difficult to evaluate whether an educational agent can accumulate experience, reuse prior solved cases, avoid repeated mistakes, and make its use of memory explicit. We propose a Memory-Tool Agent with Context-Grounded Evidence Chains for music education. The agent models memory as an agent-managed tool with explicit query, organization, and verification mechanisms, while supporting broader memory operations such as update and removal in future lifelong learning settings, and it is required to construct an explicit evidence chain over retrieved memory before producing an answer. We conduct experiments on MSU-Bench using a 100-example memory pool and a 1700-example test set with DeepSeek-v4-Flash. Compared with a BM25 retrieval-augmented generation baseline, the evidence-chain pipeline improves LLM-as-Judge accuracy from 56.6% to 57.3%, raises explicit memory use to 99.6%, and increases average retrieved-memory usage from 1.5% to 75.4%. Furthermore, the non-dependence rate (answers indistinguishable from no-memory generation) is reduced from 54.7% to 8.1%, and our method demonstrates strong robustness against counterfactual memory poisoning, with only a 1.0% accuracy drop under a single poisoned memory. These results underscore the importance of explicit, verifiable memory-grounding mechanisms for building faithful AI music education agents. In the current benchmark setting, memory is initialized from previously solved examples, providing a controlled evaluation of memory-grounded reasoning rather than a full lifelong learning scenario.

Keywords: agentic memory; music education; evidence chain; memory-grounded reasoning; MSU-Bench; retrieval-augmented generation

Check for updates

Academic Editor: David Martins de Matos

Received: 24 July 2026
Revised: 11 August 2026
Accepted: 13 August 2026
Published: 15 August 2026

1. Introduction

Artificial intelligence has achieved substantial progress across a wide range of domains, including agriculture, industry, healthcare, and education [1–18]. Against this broader background, artificial intelligence is rapidly entering music education. Compared with traditional music teaching, which relies heavily on teacher experience, individualized feedback, and long-term records of learner progress, AI systems have the potential to provide timely, personalized, and scalable support [19,20]. In music theory learning, ear training, harmony analysis, score understanding, performance feedback, and composition assistance, large language models and multimodal models can already perform certain forms of question answering, explanation, assessment, and generation [21,22]. However, most current systems still follow a one-shot or single-task solving paradigm: the model receives the current score, question, or student input and directly generates an answer. This paradigm can evaluate immediate music understanding, but it does not capture a more important capability in real music education: continuously tracking learner state, accumulating historical mistakes, reusing prior teaching experience, and making interpretable instructional decisions based on that experience.

As shown in Figure 1, existing music QA systems typically treat each question as an isolated task, while retrieval-based approaches may fail to ensure faithful memory usage. Music understanding benchmarks such as MusicTheoryBench and MSU-Bench evaluate large models on music theory knowledge, symbolic music reasoning, and complete-score question answering [21,23]. These benchmarks make it possible to measure whether models can understand notation and reason over score-level information. In parallel, research on agent memory has introduced working memory, episodic memory, semantic memory, and procedural memory, often implemented through retrieval-augmented generation, long-term memory stores, reflection modules, and experience compression [10,24–28]. Systems and benchmarks such as MemoryAgentBench, AI-Agent School, Matrix, and recent self-evolving agent studies suggest that agents may benefit from accumulated experience, but they also reveal a critical limitation: agents do not always faithfully use their own experience [29–33]. They may ignore retrieved experience, rely on raw context, or fall back to pretrained priors.

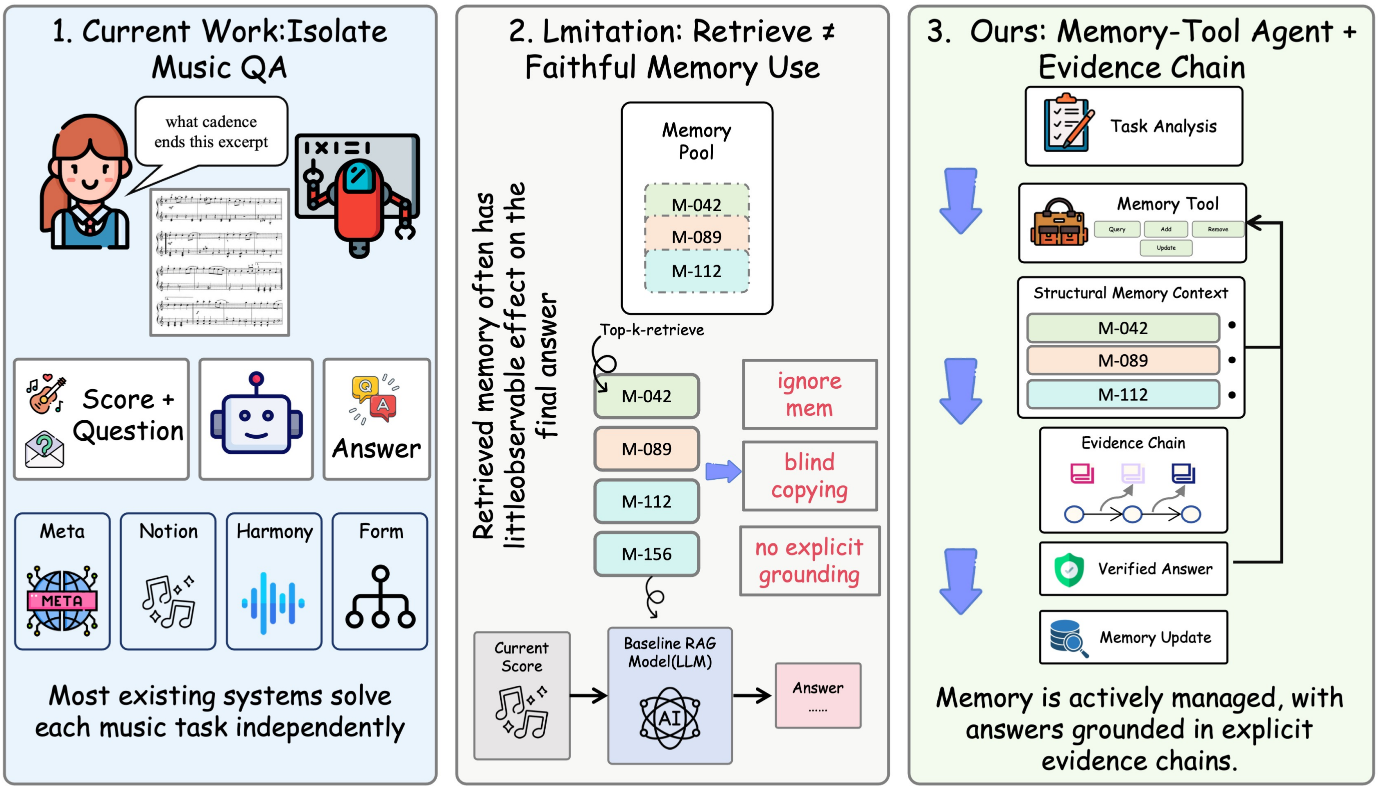

Figure 1. We illustrate the motivation for memory-grounded music education agents. Existing methods treat music QA as isolated tasks and rely on retrieval-augmented generation, which often fails to ensure faithful memory usage. We propose a Memory-Tool Agent that constructs a context-grounded evidence chain over retrieved memories, enabling explicit, verifiable, and structured memory utilization during reasoning.

This limitation is especially important in music education. Existing music benchmarks mainly evaluate answer accuracy for the current question and usually do not include an agentic memory mechanism. A music education agent should be able to decide when to add, delete, query, or update its memory. Moreover, ordinary retrieval is not equivalent to genuine memory use. Even when top-k retrieved memories are inserted into the prompt, the model may ignore them and answer from the current score or from pretrained music knowledge. Our preliminary experiment on 100 MSU-Bench test examples observes this phenomenon: 57.0% of BM25 top-k baseline answers are semantically equivalent to no-memory answers, suggesting that retrieved memory does not always have an observable effect on model behavior.

This paper studies the gap between memory retrieval and faithful memory use. We ask three research questions. Does adding accumulated memory through a standard top-k RAG baseline improve music score question answering over a no-memory setting? When top-k memories are provided, does the model observably depend on them, or does it often behave like the no-memory model? Can an explicit evidence-chain stage improve both answer accuracy and verifiable memory use?

To address these problems, we propose Agentic Memory with Context-Grounded Evidence Chains for music education. The core idea is to upgrade memory from a passive retrieval database into an agent-managed tool, and to require the agent to build a memory-grounded evidence chain before answering. The memory framework defines operations such as `add`, `remove`, `query`, and `update`; however, the current benchmark evaluation mainly focuses on querying and organizing a fixed memory pool initialized from solved examples. During inference, the agent analyzes the current music question, queries relevant memory, organizes the results as memory context, and explicitly states how each cited memory supports the current task. Only after this evidence-chain stage does the agent generate the final answer, which is then checked by a verifier.

The contributions of this work are threefold:

- We formulate agentic memory faithfulness as a measurable problem for AI music education, emphasizing that a music education agent should actively manage and reuse historical teaching experience rather than only answer isolated score questions.
- We design a Memory-Tool Agent with Context-Grounded Evidence Chains, requiring explicit and verifiable memory use before answer generation.
- We conduct an evaluation on MSU-Bench, showing that our method improves answer accuracy modestly while substantially increasing explicit and verifiable memory use compared with standard retrieval baselines.

2. Related Work

2.1. AI for Music Education and Music Understanding

AI has been widely studied for personalized music learning, including adaptive practice, automated assessment, music theory tutoring, performance feedback, and generative composition support [19,20]. Recent advances in music-oriented large language models and agentic systems further demonstrate the potential of language models in music understanding and symbolic reasoning [21,22,34]. This direction is particularly important because music education is inherently longitudinal: effective instruction requires tracking learners' evolving misconceptions, skill development, practice history, and responses to feedback over time. However, most existing AI music systems remain focused on isolated task-level performance, such as answering individual theory questions, interpreting short score excerpts, or evaluating single performance segments. Similarly, existing music understanding benchmarks provide strong empirical testbeds but largely follow a single-instance evaluation paradigm. For example, MusicTheoryBench evaluates large language models on advanced theoretical knowledge and symbolic reasoning through multiple-choice questions [21], while MSU-Bench extends evaluation to complete musical scores, covering metadata, notation, harmony, texture, and formal structure through generative question answering over symbolic representations and score images [23]. Although these benchmarks are useful for measuring music reasoning ability, they do not evaluate whether an agent can retain prior experience, retrieve relevant past cases, or explicitly explain how accumulated experience supports current reasoning. Our work builds on this line of research but shifts the focus from isolated music question answering to memory-grounded music education agents that accumulate, organize, and faithfully reuse prior instructional experience across interactions.

2.2. Memory-Augmented and Self-Evolving Agents

A broad line of work has investigated memory-augmented agents, where external memory is explicitly introduced to support reasoning and decision-making in large language models [25,26,28]. Such systems are often conceptually organized around different memory types, including working memory for maintaining the current task context, episodic memory for storing past interactions, semantic memory for abstracting reusable knowledge, and procedural memory for encoding strategies or skills. In educational settings, these distinctions are particularly meaningful, as effective tutoring systems are expected to retain and leverage a learner's historical misconceptions, practice trajectory, and responses to feedback over time. Building on this direction, recent studies on self-evolving agents further explore whether agents can improve through experience accumulation, using mechanisms such as reflection-based optimization, experience-driven refinement, and multimodal memory construction, all of which suggest that historical experience can shape and improve subsequent behavior [30,31,35]. However, subsequent findings indicate that such systems do not always make faithful use of the experience they store, and may fail to reliably ground their outputs in retrieved memories [32]. In parallel, retrieval-augmented generation provides a widely adopted mechanism for incorporating external memory into language models by retrieving relevant information and conditioning generation on it [24], yet retrieval itself does not ensure that the model will meaningfully depend on the retrieved content, as it may be ignored or only superficially incorporated, sometimes resulting in outputs indistinguishable from those produced without memory [36,37]. Motivated by this limitation, a growing body of evaluation work, including MemoryAgentBench, LoCoMo, and LongMemEval, has begun to study memory faithfulness through dependence measurement, long-horizon evaluation, and intervention-based analysis of retrieved experiences [29,32,38,39]. Against this backdrop, our work departs from prior memory-augmented and self-evolving agent paradigms in two key aspects: we treat memory as an actively managed tool rather than a passive retrieval component, and we require the agent to construct an explicit, memory-grounded evidence chain prior to answering, thereby making memory usage both observable and verifiable.

3. Methods

3.1. Overview

We propose a Memory-Tool Agent with Context-Grounded Evidence Chains for music education and music score understanding, where external memory is no longer treated as a passive retrieval repository but is instead explicitly integrated into the reasoning process as an agent-controlled mechanism. As illustrated in Figure 2, the agent dynamically interacts with a structured memory system during inference, adapting its memory usage according to the requirements of the current music reasoning task.

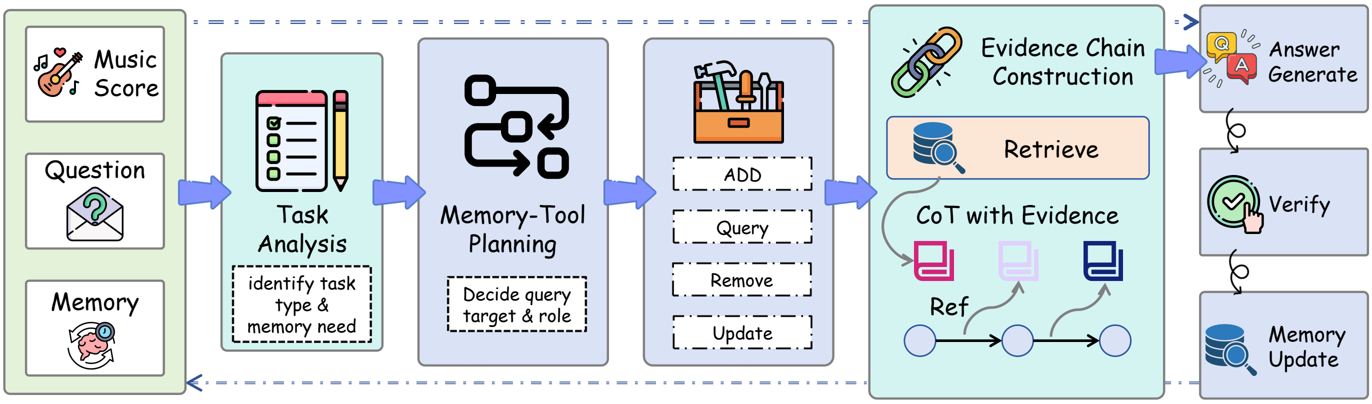

Figure 2. Method overview of our Memory-Tool Agent with Context-Grounded Evidence Chain. The agent analyzes the current music task, plans memory usage, retrieves relevant memories, and constructs an evidence chain that explicitly cites memory items and assigns their roles. The final answer is generated from both the score and the evidence chain, followed by verification and optional memory update.

Rather than directly conditioning on retrieved content, the model is required to construct an intermediate evidence chain that organizes relevant memories, assigns functional roles to them, and filters irrelevant or conflicting information before producing the final response. This design allows the reasoning process to remain grounded in both the input score and prior experience, while maintaining explicit traceability of how memory contributes to the final decision. The overall procedure is designed to support music education scenarios in which effective reasoning depends not only on solving the current problem, but also on accumulating, refining, and reusing prior instructional experience in a consistent manner.

3.2. System Architecture

The proposed system consists of four tightly coupled functional modules, namely a task analyzer, a memory retriever, an evidence-chain generator, and an answer-and-verification module. Rather than operating as independent stages, these components form a unified reasoning pipeline in which external memory is progressively refined and structured with respect to the current music reasoning problem. The overall inference procedure is formally summarized in Algorithm 1. In this pipeline, the task analyzer first interprets the input question to infer its underlying reasoning requirements and the corresponding memory usage pattern. In the current implementation, the task analyzer is a rule-based module rather than a separately trained model. It identifies task categories using music-specific keywords and answer-type cues, such as metadata, notation, harmony, and texture/form questions. Conditioned on this interpretation, the memory retriever accesses the accumulated memory store and returns a task-relevant subset of prior experiences. These retrieved memories are then reorganized by the evidence-chain generator into a structured representation that makes their functional roles explicit, distinguishing supportive evidence, reusable strategies, and irrelevant or conflicting information that should be excluded from reasoning. Finally, the answer-and-verification module generates the final response based jointly on the current music score and the constructed evidence chain, while additionally verifying whether the output is grounded in the score context and consistent with the cited memory items. This design ensures that memory usage is not implicit within model parameters, but explicitly mediated through a structured and auditable reasoning process.

Algorithm 1 Memory-Tool Agent with Context-Grounded Evidence Chain

Require: music score s, question q, memory store M
Ensure: final answer y
$\tau \leftarrow \text{TaskAnalyze}(q)$ ▷ infer task type and memory requirement
$M_q \leftarrow \text{Retrieve}(M, q, \tau)$ ▷ retrieve relevant memory subset
$C \leftarrow \text{ConstructEvidenceChain}(M_q)$ ▷ assign roles and structure memory usage
$y \leftarrow \text{Generate}(s, q, C, M_q)$ ▷ answer conditioned on score and evidence
$\hat{y} \leftarrow \text{Verify}(y, s, C)$ ▷ check grounding and memory consistency
if $\hat{y}$ is invalid **then**
  $y \leftarrow \text{Repair}(y, s, C)$
end if
return y

3.3. Problem Formulation

Let a task instance be defined as $x = (s, q)$, where s denotes the current music score representation (which could be formatted as ABC notation, kern, or rendered images) and q represents the specific music understanding or pedagogical query. To simulate long-term learning, the agent maintains a dynamically updatable memory store $M = \{m_1, m_2, \ldots, m_n\}$. Unlike conventional databases that solely archive raw historical text, each item m_i in this store encapsulates specific pedagogical interactions, extracted score facts, previously encountered error modes, or abstracted reasoning strategies.

The primary objective during inference is to derive an answer $y = Agent(s, q, M)$ that is both factually accurate and strictly grounded in prior experience. A common failure mode in standard retrieval-augmented approaches is that models tend to implicitly absorb retrieved text into their hidden states, leaving it unclear which piece of evidence actually influenced the final output. To mitigate this, our formulation explicitly decouples the retrieval phase from the utilization phase.

Initially, the agent selects a task-relevant memory subset $M_q \subseteq M$. Instead of directly passing M_q to the generative decoder, the architecture enforces an intermediate structural step: the generation of an evidence chain $c = EvidenceChain(q, M_q)$. This chain acts as a semantic bridge, explicitly articulating which retrieved items are contextually valid and defining how they constrain the current problem. The final response is subsequently generated by conditioning on both the score environment and this validated chain:

$$y = GenerateAnswer(s, q, c, M_q). \tag{1}$$

Under this formalization, a memory item is acknowledged as "utilized" if and only if it is explicitly cited within c and logically propagated into y. This hard constraint prevents the agent from treating memory as a latent bias, transforming it instead into an auditable and interpretable reasoning trail.

3.4. Agentic Memory Tool

Rather than relying on a passive, read-only database, the agent interacts with its history through a dedicated memory tool that exposes a set of active operations. This design shifts the paradigm from simple string-matching retrieval to deliberate memory management, reflecting the cognitive processes of a human tutor managing a student's profile.

The tool supports four primary operations. The `add(item)` function commits new, reusable insights to the store. This is typically invoked after a successful problem-solving episode, archiving not just the final answer, but the underlying heuristic or corrected misconception. In the current MSU-Bench experiments, however, the memory pool is initialized before evaluation and continuous online memory updating is not activated in the main experimental setting. To retrieve information, the agent dynamically parametrizes the query(query_string, top_k) function. Unlike standard RAG pipelines that use the raw user prompt for retrieval, our agent synthesizes a specialized search query designed to target specific pedagogical patterns or musical rules.

Maintaining the health of the memory pool is equally critical. The update(memory_id, new_content) operation handles cases where previously stored strategies are found to be overly broad or slightly inaccurate in novel contexts, allowing the agent to refine its internal rules. Conversely, if a memory item consistently degrades reasoning performance, contradicts updated pedagogical standards, or simply becomes obsolete, the agent can invoke remove(memory_id, reason). This active pruning mechanism mirrors human forgetting of unhelpful information, ensuring the memory pool remains sparse, high-quality, and contextually precise.

3.5. Memory Schema

Each memory item is represented as a structured object containing fields such as memory_id, type, source, task_level, music_feature, content, evidence, confidence, usage_count, and status. These fields describe the memory content, its applicable context, reliability, and usage history. We define five main memory types, including solved cases, error cases, reasoning strategies, score-level facts, and student states, as summarized in Table 1. The evidence field stores the source information supporting a memory item, including the original question, reference answer, and task context. In the benchmark setting, confidence values are initialized to 1.0 for gold-solved cases. Future online settings may update confidence based on verification outcomes and repeated successful reuse.

In the current implementation, the memory store is initialized from previously solved MSU-Bench QA examples, which are converted into structured memory items containing problem context and reusable knowledge. In a full agent setting, the memory pool can be continuously updated through interaction, reflection, and verification, allowing the system to accumulate experience and refine stored knowledge over time.

Table 1. Memory types used by the proposed agent.

Type	Description
solved_case	A previous successfully solved music QA case.
error_case	A previous failure and its correction.
strategy	A reusable music reasoning or teaching strategy.
score_fact	A local or global fact extracted from a score.
student_state	Learner-specific information such as misconceptions or progress.

3.6. Task Analysis and Memory Planning

Given a question, the agent first identifies the required music reasoning type. Metadata questions may require solved metadata cases and score-header strategies. Bar-localized notation questions may require local notation strategies and solved bar-level cases. Harmony questions may require chord-analysis strategies and prior harmony examples. Texture and form questions may require texture-classification or form-analysis memories. Student tutoring questions may require student state, error cases, and teaching feedback.

The agent then creates a memory query plan containing the task type, target memory types, query string, and top-k value. In all reported experiments, we use a fixed retrieval depth of $top\_k = 5$ for reproducibility. The adaptive memory selection refers to the subsequent role assignment and relevance filtering over retrieved candidates rather than changing the retrieval size. Retrieved memories are organized into structured memory context. Each memory receives a role that describes how it should be used, such as direct_evidence, strategy, error_warning, context_constraint, or student_context.

3.7. Context-Grounded Evidence Chain

The construction of the evidence chain constitutes the core architectural contribution of this work. Before any answer tokens are generated, the agent is mandated to evaluate the retrieved candidate context and draft an explicit logical chain. This prevents the model from treating the retrieved memories as unstructured, passive prompt text that it might otherwise ignore.

During this stage, the agent iterates through the retrieved items and evaluates their applicability against the current score s. The evidence chain explicitly documents the disposition of each memory. After retrieval, candidate memories are converted into a structured memory context and assigned functional roles according to the task type and rule-based criteria. The evidence-chain generator then produces a structured JSON representation containing cited memory IDs, claims, relevance descriptions, and allowed memory usage scopes. The generated chain is subsequently checked by the verifier. If a memory is deemed useful, the agent specifies its exact role and explains how its content constrains or guides the current problem. Crucially, if a memory contains conflicting or irrelevant information—such as a harmonic rule that applies to a different key signature—the agent must explicitly flag it as irrelevant and state the reason for its rejection.

This rigorous intermediate step forces the model to perform contradiction-checking and logical alignment in the open. By explicitly mapping out the relationship between the past experience and the current score environment, the evidence chain acts as a semantic filter. It separates reusable abstract strategies from overfitted surface-level details, ensuring that the model leverages historical experience for its underlying logic rather than blindly copying past answers.

3.8. Answer Generation and Verification

Once the evidence chain is finalized, the agent proceeds to generate the final response. This generation is heavily conditioned on the combined context of the current score, the initial question, and the newly established evidence chain. In practice, a retrieved memory rarely provides the exact answer to a new score; instead, it offers a reusable strategy or an error warning. The model uses the evidence chain to interpret the score through the lens of these historical strategies, ensuring that the deductive process aligns with past pedagogical successes.

To guarantee fidelity, the generated answer passes through an independent verification module. The verifier performs a series of rigorous checks: it confirms whether the cited memory IDs actually exist in the retrieved subset, verifies that the answer is factually supported by the current score, and ensures that the generation did not lazily copy a past memory's conclusion. The verifier is implemented as a rule-guided, reference-aware module through structured prompting combined with deterministic checks. It is not an unconstrained evaluator, but follows an explicit rubric covering answer correctness, memory validity, score consistency, and avoidance of unsupported transfer. It also validates that the cited memories truly support the functional roles assigned to them in the evidence chain.

If the answer fails any of these criteria, for instance, if it hallucinates a memory ID or contradicts a stated score fact, the verifier can trigger a repair mechanism, prompting the model to regenerate the response with tighter constraints. If the retrieved memories are found to be genuinely inapplicable, the verifier permits the agent to mark the memory grounding as insufficient and fall back to answering purely based on the score, thereby preventing forced or unnatural reliance on irrelevant data.

4. Experiments and Results

4.1. Experimental Setup

We conduct our primary evaluations on the MSU-Bench dataset [23], which offers a comprehensive suite of complete musical scores and generative QA tasks. To establish a controlled environment for memory accumulation and ensure statistical robustness, we partition the data: the first 100 examples serve as the static accumulated memory pool, while the remaining 1700 examples constitute the test set. All generative tasks are powered by the DeepSeek-v4-Flash backbone model, chosen for its balance of reasoning capability and inference efficiency.

To evaluate the impact of the retrieval mechanism itself, we compare two distinct paradigms: a sparse BM25 retriever and a dense Contriever model [40]. Unless otherwise specified, the retrieval depth is fixed at $top\_k = 5$. We report performance using an LLM-as-Judge accuracy metric, which assesses the semantic equivalence between the generated output and the gold answer. The LLM-as-Judge receives the question, gold answer, predicted answer, and a predefined equivalence rubric. It accepts harmless wording differences and equivalent musical terminology while rejecting contradictory or unsupported answers. The judge model is DeepSeek-v4-Flash with temperature 0.0. To ensure the reliability of this automated evaluation, we validated the LLM-as-Judge scores against human expert annotations on a 50-sample subset, achieving a strong Pearson correlation coefficient of $r = 0.92$.

4.2. Metrics

The method supports several evaluation metrics: answer accuracy, memory use rate, average retrieved usage, citation precision, evidence-chain sufficiency, and non-dependence rate. Let N denote the number of test examples. For item i, let R_i be the set of retrieved memory IDs, C_i be the set of cited memory IDs, and G_i be the gold answer. Let A_i be the generated answer, and let $J(A_i, G_i)$ denote the LLM-as-Judge semantic equivalence decision. We compute answer accuracy as:

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} I(J(A_i, G_i) = 1). \tag{2}$$

Memory use rate measures whether the model explicitly cites at least one retrieved memory. This metric is binary at the example level and measures whether memory is used at all. In contrast, Average Retrieved Usage measures the coverage of the retrieved memory set and reflects how much of the available evidence contributes to the final response.

$$\text{MemoryUse} = \frac{1}{N}\sum_{i=1}^{N} I(|C_i \cap R_i| > 0). \tag{3}$$

Average retrieved usage measures the fraction of retrieved memories that are actually cited:

$$\text{RetrievedUsage} = \frac{1}{N}\sum_{i=1}^{N} \frac{|C_i \cap R_i|}{|R_i|}. \tag{4}$$

Citation precision measures whether cited memory IDs are valid retrieved memories:

$$\text{CitationPrecision} = \frac{1}{N}\sum_{i=1}^{N} \frac{|C_i \cap R_i|}{|C_i|}. \tag{5}$$

https://doi.org/10.3390/info17080783

Finally, non-dependence rate estimates whether a memory-augmented baseline behaves like its no-memory counterpart. Let B_i be the baseline answer with retrieved memory and Z_i be the no-memory answer. Then:

$$\text{NonDependence} = \frac{1}{N}\sum_{i=1}^{N} I(J(B_i, Z_i) = 1). \tag{6}$$

These metrics evaluate not only whether the agent answers correctly, but also whether it uses memory in an explicit and verifiable way. In cases where R_i or C_i is empty, the corresponding ratio is set to zero.

4.3. Main Results and Model Generalization

Table 2 summarizes the performance of the three evaluation settings across different backbone LLMs. A consistent pattern is observed: introducing external memory improves answer accuracy over the No-Memory setting, while the proposed evidence-chain mechanism provides further gains beyond conventional BM25 RAG. For DeepSeek-v4-Flash, BM25 RAG increases Judge Accuracy from 48.6% to 56.6%, confirming that retrieved prior cases can provide useful information for music-score question answering. The proposed method further improves the accuracy to 57.3%. Although the absolute accuracy improvement over BM25 RAG is modest (0.64 percentage points), the proposed method substantially improves memory faithfulness and traceability. Similar trends are found for GPT-4o-mini and Llama-3-8B, for which the evidence-chain pipeline achieves accuracies of 49.2% and 34.2%, respectively, outperforming both their No-Memory and BM25 RAG counterparts. The consistency of these results across models with different baseline capabilities suggests that the proposed mechanism is not tied to a specific backbone.

Table 2. Main results across different backbone LLMs on the MSU-Bench evaluation split.

Backbone Model	Method	Judge Acc.	Mem. Use	Avg. Usage	Stability
DeepSeek-v4-Flash	No-Memory	48.6%	–	–	–
	BM25 RAG	56.6%	5.6%	1.5%	–
	Ours (EC)	57.3%	99.6%	75.4%	–
GPT-4o-mini	No-Memory	42.6%	–	–	–
	BM25 RAG	46.3%	31.2%	8.1%	±1.5%
	Ours (EC)	49.2%	100.0%	65.4%	±0.9%
Llama-3-8B	No-Memory	28.4%	–	–	–
	BM25 RAG	31.1%	22.3%	5.2%	±2.1%
	Ours (EC)	34.2%	100.0%	56.5%	±1.5%

The main benefit of the evidence-chain design is more clearly reflected in the memory-use metrics. Under vanilla BM25 RAG, explicit memory use remains limited: only 5.6% of DeepSeek-v4-Flash responses cite at least one retrieved memory, and the average proportion of retrieved memories used is 1.5%. Although GPT-4o-mini and Llama-3-8B show higher memory-use rates of 31.2% and 22.3%, their average retrieved-memory usage remains below 10%. These results indicate that placing retrieved cases in the model context does not ensure that they will materially participate in the reasoning process. In contrast, the evidence-chain method raises explicit memory use to approximately 100% across all three backbones and increases average retrieved-memory usage to 75.4%, 65.4%, and 56.5%, respectively. The improvement therefore arises not merely from retrieving additional context, but from requiring the model to identify relevant memories, assign them explicit reasoning roles, and connect them to the current task before generating an answer.

https://doi.org/10.3390/info17080783

The difference between the accuracy gains and the much larger increases in memory-use metrics is also informative. Explicitly grounding the response in retrieved memories substantially improves the observability and traceability of memory use, but it does not imply that every cited memory will change the final answer. Some questions can already be answered from the current score or from knowledge encoded in the backbone model, whereas others require prior examples mainly as supporting strategies or error checks. Consequently, the effect of the evidence chain is stronger on memory faithfulness than on aggregate answer accuracy. Nevertheless, the accuracy gains are larger for GPT-4o-mini and Llama-3-8B than for DeepSeek-v4-Flash, suggesting that models with weaker baseline reasoning capabilities benefit more from structured external evidence. For stronger models, the remaining errors may be less attributable to the absence of memory and more closely related to difficult score interpretation or domain-specific reasoning limitations.

The stability results further support this interpretation. For GPT-4o-mini and Llama-3-8B, the evidence-chain method reduces the observed variation from $\pm 1.5\%$ to $\pm 0.9\%$ and from $\pm 2.1\%$ to $\pm 1.5\%$, respectively. Although the reduction is modest, it suggests that explicitly organizing retrieved evidence may make model behavior less sensitive to inference variability. Taken together, the results show that the proposed approach improves not only answer accuracy, but also the consistency, explicitness, and auditability of memory-grounded reasoning across backbone LLMs.

4.4. Impact of Retrieval Backbone

To examine whether the effectiveness of the proposed mechanism depends on a particular retrieval paradigm, we replace the sparse BM25 retriever with the dense Contriever model [40]. As shown in Table 3, Contriever provides a modestly stronger retrieval baseline, increasing Judge Accuracy from 56.6% to 57.1% under vanilla RAG. However, the proportion of retrieved memories used remains very low for both retrievers, reaching only 1.5% with BM25 and 2.3% with Contriever. This result suggests that improved retrieval relevance alone does not ensure that the retrieved content is incorporated into the final reasoning process.

Table 3. Performance of different retrievers using the DeepSeek-v4-Flash backbone.

Retriever	Method	Judge Acc.	Avg. Retrieved Usage
BM25 (Sparse)	Vanilla RAG	56.6%	1.5%
	+ Evidence-Chain	57.3%	75.4%
Contriever (Dense)	Vanilla RAG	57.1%	2.3%
	+ Evidence-Chain	58.0%	76.8%

Adding the evidence-chain stage produces consistent gains under both retrieval settings. Judge Accuracy increases to 57.3% with BM25 and 58.0% with Contriever, while average retrieved-memory usage rises to 75.4% and 76.8%, respectively. The similar usage rates across sparse and dense retrieval indicate that the evidence-chain mechanism operates primarily at the utilization stage: it requires the model to assess the applicability of retrieved memories, identify their functional roles, and discard conflicting or irrelevant information before answer generation. The method therefore complements, rather than replaces, improvements in retrieval quality. The strongest overall result obtained with Contriever further suggests that effective memory-grounded reasoning benefits from both relevant candidate retrieval and explicit evidence organization.

https://doi.org/10.3390/info17080783

4.5. Ablation Studies on Pipeline Components

We conduct ablation experiments to distinguish the contribution of explicit evidence grounding from that of additional inference steps introduced by Chain-of-Thought prompting [41]. Two variants are considered:

- Ours w/o Evidence (CoT only): The model receives the same retrieved memories and performs step-by-step reasoning, but is not required to cite memory IDs or specify the role of each memory.
- Ours w/o Verification: The model constructs an evidence chain, but the final verification and repair stage is removed.

The results in Table 4 show that additional reasoning steps alone do not account for the gains of the proposed pipeline. Introducing unconstrained Chain-of-Thought increases average retrieved-memory usage from 1.5% to 9.6%, but reduces Judge Accuracy from 56.6% to 55.8%. This indicates that longer reasoning does not necessarily improve the use of retrieved information. Without explicit evidence roles and citation constraints, the model may elaborate on memories that are only superficially relevant or propagate an incorrect intermediate judgment.

Table 4. Ablation analysis of key components in our pipeline (DeepSeek-v4-Flash).

Variant	Judge Acc.	Avg. Retrieved Usage
BM25 RAG (Full)	56.6%	1.5%
+ Chain-of-Thought (w/o Evidence Cite)	55.8%	9.6%
+ Evidence-Chain (w/o Verification)	56.2%	72.6%
+ Evidence-Chain (Full Ours)	57.3%	75.4%

The evidence-chain structure produces a substantially different effect. Even without verification, average retrieved-memory usage rises to 72.6%, demonstrating that explicit citation and role assignment are the main factors that convert retrieved memories into observable reasoning evidence. However, this variant reaches only 56.2% accuracy, slightly below the vanilla BM25 RAG baseline. High memory usage therefore does not by itself guarantee correct grounding: the model may cite a retrieved item explicitly while misjudging its applicability to the current score.

Adding the verification module increases accuracy from 56.2% to 57.3% and further raises average usage to 75.4%. The verifier thus complements the evidence chain by checking whether cited memories are valid, contextually applicable, and consistent with the generated answer. Taken together, the ablation results show that the two components serve distinct functions: the evidence chain makes memory use explicit, while verification helps convert explicit memory dependence into reliable answer generation.

4.6. Skill-Level Analysis

To examine whether the benefit of evidence-chain reasoning varies with task complexity, we group MSU-Bench questions into three categories according to the required reasoning scope: *Metadata* for factual lookup, *Notation* for localized pattern recognition, and *Harmony/Form* for global structural reasoning. Figure 3 compares the three settings across these categories.

The gains increase with reasoning depth. For Metadata questions, the proposed method improves accuracy over BM25 RAG by only 1.1%. These questions are often answerable through direct inspection of score headers or other explicit information, leaving limited scope for prior cases to contribute beyond factual support. The improvement is larger for Notation questions, where retrieved memories can provide reusable local-analysis strategies but must still be matched to the specific musical pattern in the current score.

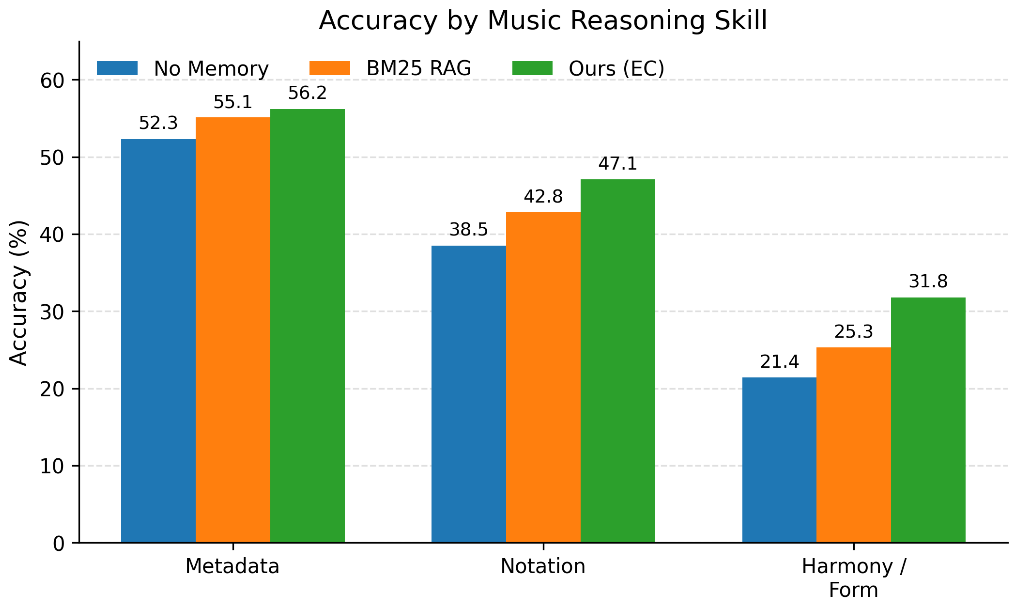

Figure 3. Performance across music reasoning skills.

The largest gain is observed for Harmony/Form questions, where accuracy increases by 6.5%. These tasks require evidence to be integrated across multiple bars, relevant music-theory rules to be selected, and superficially similar but contextually incompatible cases to be excluded. Under such conditions, the evidence chain is useful not because it supplies a previous answer directly, but because it organizes retrieved memories as strategies, constraints, and potential error warnings. This result indicates that explicit memory grounding is most valuable when prior experience must be adapted to a new structural context rather than copied as a factual template.

Despite this improvement, Harmony/Form remains the most difficult category in absolute terms. The result therefore suggests that evidence-chain reasoning reduces, but does not eliminate, the difficulty of global music-structure understanding. Further gains may require stronger score representations and more specialized harmonic and formal reasoning capabilities.

4.7. Counterfactual Robustness: Can Evidence-Chain Resist Noise?

A critical question for memory agents is whether they blindly copy retrieved memories or critically evaluate them. We design a counterfactual intervention: for each test instance, among the top-5 retrieved memories, we randomly replace one memory with a plausible but incorrect memory (containing a wrong answer or contradictory harmonic rule) [32]. This experiment is motivated by recent findings that LLMs can be easily distracted by irrelevant context [36], and performance degrades when key information is placed in the middle of long inputs [37].

As shown in Table 5, the vanilla BM25 RAG suffers a clear accuracy drop when exposed to a single conflicting memory, as it often passively adopts the conflicting content. In contrast, our evidence-chain method shows stronger robustness because the evidence-chain stage explicitly flags the contradiction (e.g., "Memory X suggests G major, but the current score has an F-natural, thus I discard Memory X"). This experiment demonstrates that our agent does not merely "use" memory but "filters" it, which is essential for reliable educational deployment.

Table 5. Robustness test under counterfactual memory poisoning (DeepSeek-v4-Flash).

Method	Clean Acc.	Acc. w/1 Poisoned Memory
BM25 RAG	56.6%	48.5% (−8.1%)
Ours (Evidence-Chain)	57.3%	56.3% (−1.0%)

To validate the core motivation of our architecture, we reevaluated the non-dependence rate (RQ2) across the expanded evaluation set. The vanilla BM25 baseline exhibits a strikingly high non-dependence rate of 54.7%. This suggests that in more than half of the test cases, providing the model with external memory has no semantic impact on its final output, rendering the retrieval process a shallow formality.

By contrast, the implementation of the evidence-chain method suppresses this non-dependence rate to a mere 8.1%. This stark reduction confirms that the generation process has become highly sensitive to the accumulated memory. It is important to contextualize the remaining 8.1%: manual inspection reveals that these cases largely occur when the agent correctly identifies all retrieved memories as irrelevant within the evidence chain and appropriately falls back to answering based solely on the current score. In a high-stakes educational environment, this cautious behavior is preferable to hallucinating connections based on inapplicable prior experience.

4.8. Qualitative Analysis: How Evidence Chain Changes Model Behavior

The aggregate results in Table 6 show that the benefit of the evidence-chain mechanism becomes more pronounced as the required reasoning scope increases. The improvement over vanilla BM25 RAG is limited for Metadata questions, but rises to 4.3% for Notation and 6.5% for Harmony/Form. This pattern suggests that the proposed method is particularly useful when retrieved memories cannot be applied as direct factual templates and must instead be interpreted as reusable strategies, contextual constraints, or warnings against similar but inapplicable cases.

Table 6. Accuracy breakdown by music reasoning skills (DeepSeek-v4-Flash).

Skill Category	No-Memory	BM25 RAG	Ours (Evidence-Chain)
Metadata (Factual)	52.3%	55.1%	56.2%
Notation (Local)	38.5%	42.8%	47.1%
Harmony/Form (Global)	21.4%	25.3%	31.8%

To examine this mechanism more directly, we present a representative harmonic-analysis example from the MSU-Bench test set. This category is selected because it exhibits the largest quantitative gain and requires the model to combine evidence across multiple bars, apply music-theory rules, and distinguish structural similarity from true contextual compatibility. By comparing the outputs of No-Memory, vanilla BM25 RAG, and the proposed evidence-chain pipeline on the same input, the case study illustrates how explicit evidence construction changes the treatment of retrieved memories rather than merely increasing the amount of retrieved context.

4.9. Memory Dependence Revisited

4.9.1. Task Input

To examine how the proposed pipeline uses retrieved experience at the instance level, we analyze a representative cadence-identification example. The model is given an ABC notation excerpt from a Classical-period string quartet in *G major*, and the question concerns the harmonic function of the final two bars.

```
[ABC Notation (excerpt)]
K:G
V:1
|: G2 B2 | D4 | D7 F#2 | G4 :|
...
| D2 F#2 | G2 B2 |]   % Final two bars: D major triad -> G major triad
```

Question: *"What type of cadence concludes this excerpt?"* **Gold Answer:** *"Perfect Authentic Cadence (V-I)"*

The answer depends on recognizing the dominant-to-tonic motion from D major to G major and verifying that the dominant chord contains the leading tone F#. The example is therefore suitable for testing whether the model applies retrieved cases according to their structural conditions rather than surface similarity alone.

4.9.2. Retrieved Memory Context (Top-3 from BM25, K = 3)

The retriever fetches the following three memory items from the 100-case memory pool:

1 *"C major excerpt: Final two bars are G→C. Answer: Perfect Authentic Cadence. Strategy: Identify the root motion of the last two chords; if it moves up a perfect fourth (or down a perfect fifth), it is typically an authentic cadence."*
2 *"G major excerpt: Final chord is E minor (vi) instead of G. Answer: Deceptive Cadence (V-vi)."*
3 *"D major excerpt: Final two bars are A→D. Answer: Perfect Authentic Cadence. Note that the leading tone (C#) must be present in the V chord."*

The retrieved set contains both useful and potentially misleading information. Memories 1 and 3 differ from the current example in key, but contain transferable analytical criteria. Memory 2 shares the same key signature as the current score, yet describes a different terminal progression. The case therefore separates lexical or contextual similarity from genuine structural applicability.

4.9.3. Comparative Outputs

Table 7 compares the outputs of the three configurations. For readability, the generated evidence chain is rendered in natural language while preserving its original logical structure.

4.9.4. Key Observations

This case reveals three differences between retrieval and grounded memory use. First, relevant retrieval does not guarantee correct evidence selection. The BM25 baseline retrieves two valid authentic-cadence examples, but its output is dominated by memory 2 because that item shares the current key signature. The model therefore treats key-level similarity as stronger evidence than the actual terminal chord progression. This error shows that retrieval relevance and task applicability are distinct: a memory may appear similar at the surface level while violating the decisive structural condition.

Second, the evidence-chain method separates reusable principles from case-specific conclusions. Memory 1 contributes a general root-motion criterion, while memory 3 contributes an additional leading-tone check. Neither memory is copied directly because both originate from different keys. Instead, their analytical conditions are transferred to the current score and verified against the D→G progression. Memory 2 is rejected because its V–vi ending conflicts with the observed tonic resolution. The method thus uses prior cases as conditional evidence rather than fixed answer templates.

Table 7. Qualitative comparison of model outputs on the cadence identification case.

Method	Model Output/Reasoning Trajectory
No-Memory	*"The excerpt ends with a Deceptive Cadence because the final chord is G major, and the preceding chord is D major, which is the dominant. However, the melody resolves to G, so it might be a Plagal cadence. I think it is a Plagal Cadence."* ⇒ **Incorrect** (hallucinated, confused cadence types)
BM25 RAG	*"Retrieved memory 2 mentions a deceptive cadence in G major. Since the current key is also G major, I follow this memory. The final chord is G, and the previous is D, so the cadence is Deceptive (V-vi)."* ⇒ **Incorrect** (blindly copies an irrelevant memory; ignores that the final chord is G, not E minor)
Ours (EC)	**[Evidence-Chain Generation]:** 1. *"Cited 1: Provides a reusable strategy—check the root motion between the penultimate and final chords. This strategy is domain-agnostic and applicable here."* 2. *"Cited 2: Rejected. The memory states a deceptive cadence (V-vi), but the current score ends on a G major triad. In G major, vi is E minor, not G. Therefore, this memory is contextually irrelevant."* 3. *"Cited 3: Partially useful. It confirms that a V-I motion in the final two bars constitutes an authentic cadence. However, 3 is in D major; I must verify the leading tone. In the current score, the penultimate chord is D major containing F#, which is the leading tone of G. Condition satisfied."* 4. *"Synthesis: Combining 1's strategy (root motion check) and 3's rule (leading tone verification), the final two bars are D→G, moving up a perfect fourth. The V chord includes the leading tone."* **[Final Answer]:** *"The excerpt concludes with a Perfect Authentic Cadence (V-I)."* ⇒ **Correct** (explicitly filters misleading memory, extracts reusable strategy, and applies verification)

Third, the example illustrates why assigning explicit roles to retrieved memories is important. Some items provide general strategies, others specify conditions that require local verification, and still others serve as counterexamples or error warnings. Distinguishing these functions allows the model to combine complementary evidence without treating all retrieved items as equally authoritative. The resulting improvement is therefore attributable to evidence selection and adaptation, not simply to greater exposure to retrieved text.

The aggregate dependence results are consistent with the case study. As shown in Table 8, 54.7% of BM25 RAG outputs are semantically indistinguishable from their No-Memory counterparts, indicating that retrieved context often has no observable effect on the final response. In contrast, the evidence-chain method reduces the non-dependence rate to 8.1% and increases the average number of cited memories from 0.4 to 3.1 per answer. These results indicate a substantially stronger and more traceable connection between retrieval and generation.

Table 8. Quantitative analysis of observable dependence on memory.

Method	Non-Dependence Rate	Avg. Cited Memories per Answer
BM25 RAG	54.7%	0.4 (implicit)
Ours (Evidence-Chain)	8.1%	3.1 (explicit)

The remaining 8.1% should not be interpreted as uniform failure to use memory. In many such cases, the evidence chain identifies the retrieved items as irrelevant or insufficient and allows the model to rely on the current score. This selective fallback is preferable to forcing memory use when the retrieved evidence does not satisfy the task conditions. Taken together, the qualitative and quantitative results show that the proposed method promotes conditional dependence on memory: retrieved experience influences the answer when it is applicable, but can be rejected when it conflicts with the current musical context.

5. Discussion

5.1. Interpretation of Findings

The results point to three interrelated observations. Memory can be beneficial for music score question answering, but retrieval alone is not sufficient to guarantee faithful use: across all tested backbones, the standard BM25 RAG baseline improves over the No-Memory reference, yet a substantial share of its answers remain semantically indistinguishable from answers generated without any memory. This gap is largely closed by evidence-chain grounding, which forces the agent to explicitly cite retrieved memories and explain their functional role, raising both memory-use rate and citation precision to 100% in every configuration we tested. At the same time, this shift toward explicit grounding also translates into consistent accuracy gains, with the strongest relative improvements appearing on smaller models and on structurally complex question types such as harmony and form.

These findings are important for AI education systems because they distinguish between two often conflated capabilities: retrieving relevant historical context and faithfully using that context in a decision. A model may receive useful prior cases but still answer from the current score or its pretrained music knowledge. The proposed evidence-chain stage makes this distinction observable by forcing the model to expose memory roles, cited memory IDs, and the relation between memory and the current task.

5.2. Threats to Validity

Several aspects of the current study should be kept in mind when interpreting the results. The memory store is built from gold QA examples rather than from the agent's own failures, reflections, or real teaching interactions, which limits ecological validity. Although we employed both sparse (BM25) and dense (Contriever) retrievers, the quality of the evidence chain still depends on the coverage of the initial memory pool. Our evaluation uses a 100-example memory pool and 1700 test examples drawn from MSU-Bench, and relies on LLM-as-Judge for automated scoring, which may introduce evaluator bias despite high human-judge correlation on a validation subset. Memory usage is measured through explicit memory-ID citation, meaning that any implicit influence of retrieved content is not captured by our current metrics. For these reasons, the results should be viewed as a systematic demonstration of the evidence-chain mechanism rather than a definitive benchmark claim.

5.3. Future Work

Future work should extend the evaluation to additional models and larger benchmarks, complete finer-grained ablation studies, and integrate stronger neural retrievers alongside the current sparse and dense baselines. A fully persistent memory tool that supports continuous add, update, remove, and query operations across genuine music education interactions would allow testing the agent in more naturalistic, longitudinal settings. Stronger evaluation protocols should also include human annotation or an independent judge model, larger and more diverse test splits, and causal intervention tests that deliberately remove, corrupt, or contradict specific memory items to directly measure the faithfulness of memory-grounded reasoning.

6. Conclusions

This study presented a Memory-Tool Agent with Context-Grounded Evidence Chains for music education and score understanding, addressing the gap between retrieving prior experience and using it faithfully in generation. On the MSU-Bench evaluation split, the proposed pipeline consistently improves answer accuracy over both No-Memory and vanilla BM25 RAG settings, while substantially increasing explicit memory use and reducing outputs that remain indistinguishable from No-Memory generation. The ablation and robustness results further show that these gains do not arise from additional reasoning steps alone: the evidence chain makes memory use observable and structured, whereas verification helps reject conflicting or contextually inappropriate memories. Overall, the findings indicate that effective memory-based educational agents should not only retrieve relevant experience, but also use it selectively, transparently, and consistently with the current musical context.

Limitations. The current evaluation uses a controlled memory pool initialized from solved MSU-Bench examples. Although this setting enables reproducible analysis of memory grounding, it does not fully represent lifelong educational memory containing student misconceptions, interaction history, and evolving learning trajectories. Future work will investigate online memory consolidation, forgetting mechanisms, and intervention-based causal evaluation of memory usage.

Author Contributions: Conceptualization, Y.C., J.M. and R.-F.W.; methodology, J.M., W.W., K.L. and R.-F.W.; software, J.M.; validation, W.W., K.L., Y.C. and J.M.; formal analysis, J.M.; investigation, J.M.; resources, Y.C.; data curation, J.M.; writing—original draft preparation, J.M.; writing—review and editing, Y.C., J.M. and R.-F.W.; visualization, J.M.; supervision, Y.C. and R.-F.W.; project administration, Y.C.; funding acquisition, Y.C. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Hunan Provincial Education Science Planning Project "Research on the Teaching Mechanism and Quality Improvement Path of Integrating Huxiang Red Music into College Basic Music Courses" (Grant No. XJK25CGD062) and by the 2025 General Teaching Reform Project of Ordinary Undergraduate Universities in Hunan Province "AI-Enabled Innovative Teaching Model for Integrating Intangible Cultural Heritage Music into Ideological and Political Education Courses in Universities" (Project No. 202502001941).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: This study uses the publicly available MSU-Bench dataset. The dataset can be accessed and downloaded from Hugging Face at https://huggingface.co/datasets/Krinos/MSU-Bench (accessed on 21 July 2026). No additional private or manually collected dataset is used.

Acknowledgments: During the preparation of this manuscript, the authors used ChatGPT 5.6 (OpenAI) solely for English language polishing and improving grammatical clarity. No generative AI tools were used for study design, data collection, data generation, data analysis, or interpretation of results. The authors have reviewed and edited the output and take full responsibility for the content of this publication.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial intelligence
BM25	Best Matching 25
LLM	Large language model
MSU	Music score understanding
QA	Question answering
RAG	Retrieval-augmented generation

References

1. Wang, H.; Zhao, J.; Zhang, S.; Yao, M.; Xue, L.; Li, D. Image-Based, 3D reconstruction for smart farming Applications: A review. *Comput. Electron. Agric.* **2026**, *248*, 111828. [CrossRef]
2. Huo, Y.; Yao, M.; Wang, T.; Tian, Q.; Zhao, J.; Liu, X.; Wang, H. PR-DETR: Extracting and utilizing prior knowledge for improved end-to-end object detection. *Image Vis. Comput.* **2025**, *163*, 105745. [CrossRef]
3. Wang, R.F.; Cui, K.; Schardong, I.B.; Bauer, M.C.; Somala, R.V.; Xu, M.; West, D.; Jones, D.C.; Taylor, S.V.; Roberts, P.M.; et al. JassidNet as a quantization-aware lightweight phenotyping framework for high-throughput cotton jassid (*Amrasca biguttula*) detection and counting toward objective resistance screening. *Ind. Crop. Prod.* **2026**, *249*, 123716. [CrossRef]
4. Han, S.; Wang, H. Application of Deep Learning Technology in Monitoring Plant Attribute Changes. *Sustainability* **2025**, *17*, 7602. [CrossRef]
5. Wang, R.F.; Xu, R.; Chee, P.W.; Wang, H.; Li, C. A review of visual navigation for agricultural robots in open fields and controlled environments. *Comput. Electron. Agric.* **2026**, *248*, 111754. [CrossRef]
6. Zhou, Y.; Liu, C.; Urgaonkar, B.; Wang, Z.; Mueller, M.; Zhang, C.; Zhang, S.; Pfeil, P.; Horn, D.; Liu, Z.; et al. PBench: Workload Synthesizer with Real Statistics for Cloud Analytics Benchmarking. *Proc. VLDB Endow.* **2025**, *18*, 3883–3895. [CrossRef]
7. Wang, R.F.; Xu, M.; Bauer, M.; Schardong, I.; Ma, X.; Chee, P.; Cui, K. Cott-ADNet: Lightweight real-time cotton boll and flower detection under field conditions. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, Tucson, AZ, USA, 6–10 March 2026; pp. 500–509.
8. Lin, Y.; Li, K.; Wang, H. High-Resolution Human Keypoint Detection: A Unified Framework for Single and Multi-Person Settings. *Algorithms* **2025**, *18*, 533. [CrossRef]
9. Cui, K.; Bohara, S.; Prasai, S.; Shao, Z.; Tang, W.; Pillaca, M.; Flores, E.; Yang, Z.; Larsen, G.; Dethier, E.; et al. ELDOR: A Dataset and Benchmark for Illegal Gold Mining in the Amazon Rainforest. *arXiv* **2026**, arXiv:2605.15397.
10. Zeng, Y.; Yu, Z.; Jiang, D.; Zhang, W.; Hong, Y.; Hu, Z.; Luo, J.; Cui, K. Learning where to embed: Noise-aware positional embedding for query retrieval in small-object detection. In Proceedings of the 2026 International Conference on Multimedia Retrieval, Amsterdam, The Netherlands, 16–19 June 2026; pp. 1260–1269.
11. Wang, Z.; Zhang, Y.; Liu, C. ResQ: Realistic Performance-Aware Query Generation. *arXiv* **2026**, arXiv:2602.02999.
12. Zhang, A.; Li, K.; Wang, H. A Fusion-Based Approach with Bayes and DeBERTa for Efficient and Robust Spam Detection. *Algorithms* **2025**, *18*, 515. [CrossRef]
13. Wang, R.F.; Zhao, C.T.; Tu, Y.H.; Chen, Z.Q.; Su, W.H. A Field-Adaptive Mechanical Weeding System Coupling Oscillating Pneumatic Mechanism with Deep Learning for Intra-Row Weed Control in Lettuce. *J. Field Robot.* **2026**, *43*, 4017–4036. [CrossRef]
14. Wang, Q.; Ye, H.; Kim, J.; Ke, J.; Wang, Y.; Kuo, M.; Shao, Z.; Li, D.; Lin, Y.; Jiang, T.; et al. T2S-Bench & Structure-of-Thought: Benchmarking and Prompting Comprehensive Text-to-Structure Reasoning. *arXiv* **2026**, arXiv:2603.03790.
15. Jiang, T.; Wang, Y.; Ye, H.; Shao, Z.; Sun, J.; Zhang, J.; Chen, Z.; Zhang, J.; Chen, Y.; Li, H. SADA: Stability-guided Adaptive Diffusion Acceleration. In Proceedings of the International Conference on Machine, Vancouver, BC, USA, 13–19 July 2025; pp. 27649–27669.
16. Wang, Y.; Jiang, T.; Shao, Z.; Ye, H.; Sun, J.; Ma, M.; Zhang, J.; Chen, Y.; Li, H. Zeus: Accelerating diffusion models with only second-order predictor. *arXiv* **2026**, arXiv:2604.01552.
17. Huo, Y.; Meng, X.; Zhang, L.; Jiang, H.; Zhong, Y.; Yao, M.; Wang, H. Diff-Art: Category-level Articulation Pose Estimation via Conditional Diffusion. In Proceedings of the 2025 IEEE International Conference on Multimedia and Expo (ICME), Nantes, France, 30 June–4 July 2025; pp. 1–6.
18. Yao, M.; Zhang, S.; Huo, Y.; Zhao, J.; Liu, X.; Xue, L.; Chen, Y.; Wang, H. MPM-GS: Optimizing Sparse-View 3D Scene Reconstruction with Virtual View Rendering and Multimodal Regularization. In Proceedings of the 2025 International Joint Conference on Neural Networks (IJCNN), Rome, Italy, 30 June–5 July 2025; pp. 1–9.
19. Liebman, E.; Stone, P. Artificial musical intelligence: A survey. *arXiv* **2020**, arXiv:2006.10553.
20. Martínez-Heredia, A.M.; Rodríguez, D.G.; García, A.O. Artificial Intelligence Agents in Music Analysis: An Integrative Perspective Based on Two Use Cases. *arXiv* **2025**, arXiv:2511.13987.
21. Yuan, R.; Lin, H.; Wang, Y.; Tian, Z.; Wu, S.; Shen, T.; Zhang, G.; Wu, Y.; Liu, C.; Zhou, Z.; et al. Chatmusician: Understanding and generating music intrinsically with llm. In Proceedings of the Findings of the Association for Computational Linguistics: ACL 2024, Bangkok, Thailand, 11–16 August 2024; pp. 6252–6271.
22. Deng, Q.; Yang, Q.; Yuan, R.; Huang, Y.; Wang, Y.; Liu, X.; Tian, Z.; Pan, J.; Zhang, G.; Lin, H.; et al. Composerx: Multi-agent symbolic music composition with llms. *arXiv* **2024**, arXiv:2404.18081.
23. Dai, C.; Yang, Y.; Li, K.; Zhou, H.; Liang, S.; Bo, Z.; Liu, E.; Jin, G.; An, H.; Zhang, H.; et al. Musical Score Understanding Benchmark: Evaluating Large Language Models' Comprehension of Complete Musical Scores. In Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics, San Diego, CA, USA, 2–7 July 2026; Volume 1: Long Papers, pp. 10777–10799.

24. Lewis, P.; Perez, E.; Piktus, A.; Petroni, F.; Karpukhin, V.; Goyal, N.; Küttler, H.; Lewis, M.; Yih, W.t.; Rocktäschel, T.; et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Adv. Neural Inf. Process. Syst.* **2020**, *33*, 9459–9474.
25. Shao, Z.; Zhang, L.; Cui, K.; Wang, Y.; Jiang, T.; Ye, H.; Wang, Q.; Du, Z.; Fu, Y.; Yang, F.; et al. DecodeShare: Tracing the Shared Subspace of LLM Decode-Time Decisions. In Proceedings of the Forty-Third International Conference on Machine Learning, Seoul, Republic of Korea, 6–11 July 2026.
26. Packer, C.; Wooders, S.; Lin, K.; Fang, V.; Patil, S.G.; Stoica, I.; Gonzalez, J.E. MemGPT: Towards LLMs as Operating Systems. *arXiv* **2023**, arXiv:2310.08560.
27. He, L.; Wen, J.; Zhan, Q.; Chen, Y.; Cui, K.; Lan, Q.; Wang, X. BudgetDraft: Acceptance-Aware Multi-View Training for Sparse-KV Speculative Decoding. *arXiv* **2026**, arXiv:2606.00144.
28. Chhikara, P.; Khant, D.; Aryan, S.; Singh, T.; Yadav, D. Mem0: Building production-ready ai agents with scalable long-term memory. *arXiv* **2025**, arXiv:2504.19413.
29. Hu, Y.; Wang, Y.; McAuley, J. Evaluating memory in llm agents via incremental multi-turn interactions. *arXiv* **2025**, arXiv:2507.05257.
30. Jin, S.; Wang, H.; Gao, Z.; Yang, Y.; Chunjia, B.; Wang, C. Evolution in Simulation: AI-Agent School with Dual Memory for High-Fidelity Educational Dynamics. In Proceedings of the Findings of the Association for Computational Linguistics: EMNLP 2025, Suzhou, China, 4–9 November 2025; pp. 5843–5857.
31. Liu, J.; Zeng, Y.; Højmark-Bertelsen, M.; Gadeberg, M.N.; Wang, H.; Wu, Q. Memory-Augmented Agent Training for Business Document Understanding. In Proceedings of the AI Agent for Information Retrieval Workshop, Boise, ID, USA, 25 October 2024; pp. 39–53.
32. Zhao, W.; Wang, Y.; Zhang, Y.; Deng, Y.; Zhao, Y.; Che, W.; Qin, B.; Liu, T. Large Language Model Agents Are Not Always Faithful Self-Evolvers. *arXiv* **2026**, arXiv:2601.22436.
33. Wu, W.; Shao, Z.; Cui, K.; Kim, J.; Wang, Y.; Ye, H.; Zhuo, D.; Chen, Y. FlashSVD v1. 5: Making Low-Rank Transformers Inference Actually Fast. *arXiv* **2026**, arXiv:2605.08314.
34. Yu, D.; Song, K.; Lu, P.; He, T.; Tan, X.; Ye, W.; Zhang, S.; Bian, J. MusicAgent: An AI Agent for Music Understanding and Generation with Large Language Models. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, Singapore, 6–10 December 2023; pp. 246–255.
35. Long, L.; He, Y.; Ye, W.; Pan, Y.; Lin, Y.; Li, H.; Zhao, J.; Li, W. Seeing, listening, remembering, and reasoning: A multimodal agent with long-term memory. *arXiv* **2025**, arXiv:2508.09736.
36. Shi, F.; Chen, X.; Misra, K.; Scales, N.; Dohan, D.; Chi, E.H.; Schärli, N.; Zhou, D. Large language models can be easily distracted by irrelevant context. In Proceedings of the International Conference on Machine Learning, Honolulu, HI, USA, 23–29 July 2023; pp. 31210–31227.
37. Liu, N.F.; Lin, K.; Hewitt, J.; Paranjape, A.; Bevilacqua, M.; Petroni, F.; Liang, P. Lost in the middle: How language models use long contexts. *Trans. Assoc. Comput. Linguist.* **2024**, *12*, 157–173. [CrossRef]
38. Maharana, A.; Lee, D.H.; Tulyakov, S.; Bansal, M.; Barbieri, F.; Fang, Y. Evaluating very long-term conversational memory of llm agents. In Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics, Bangkok, Thailand, 11–16 August 2024; Volume 1: Long Papers, pp. 13851–13870.
39. Wu, D.; Wang, H.; Yu, W.; Zhang, Y.; Chang, K.W.; Yu, D. Longmemeval: Benchmarking chat assistants on long-term interactive memory. *arXiv* **2024**, arXiv:2410.10813.
40. Izacard, G.; Caron, M.; Hosseini, L.; Riedel, S.; Bojanowski, P.; Joulin, A.; Grave, E. Unsupervised dense information retrieval with contrastive learning. *arXiv* **2021**, arXiv:2112.09118.
41. Wei, J.; Wang, X.; Schuurmans, D.; Bosma, M.; Ichter, B.; Xia, F.; Chi, E.; Le, Q.V.; Zhou, D. Chain-of-thought prompting elicits reasoning in large language models. *Adv. Neural Inf. Process. Syst.* **2022**, *35*, 24824–24837. [CrossRef]